Palaeomagnetic data on the andesitic cover of the Sesia-Lanzo Zone (Western Alps)

By Roberto Lanza, Turin *)

With 4 figures and 2 tables.

Zusammenfassung

Die natürliche remanente Magnetisierung der Andesiten, die die Bedeckung der Sesia-Lanzo Zone bilden, ist an 16 Orten gemessen worden, die entlang drei Querdurchschnitten liegen. Das Gestein besitzt eine stabile NRM, die auf Grund eines Konglomeratstest die ursprüngliche sein müßte. Alle Orte, die annehmbare statistische Werte aufweisen, außer einem, haben die gleiche Magnetisierungsrichtung. Die paläomagnetischen Messungen zeigen keine Strukturen aufeinanderfolgender Deformation bei dem Magnetisierungsvorgang. Die durchschnittliche Magnetisierungsrichtung (11 Orte, 152 Gesteinsproben) ist: D = 135,9, I = —2,9, mit α_{95} = 8,8. Keine tektonische Korrektur wird bisher eingerechnet.

Abstract

The remanent magnetization of the andesitic cover of the Sesia-Lanzo Zone has been studied from 16 sites along three transversal sections. The remanent magnetization of the rock is stable and it appears to be primary after a conglomerate test. The mean direction of magnetization is consistent for all sites, but for one exception, with satisfactory values of statistical parameters. Therefore no deformation structures postdating the acquisition of remanent magnetization can be inferred from palaeomagnetic data. The mean direction of magnetization (11 sites, 152 specimens) is: D = 135.9, I = —2.9, with α_{95} = 8.8. No tectonic correction can yet be made.

Résumé

L'aimantation rémanente des andésites qui constituent la couverture de la Zone Sesia-Lanzo a été mesurée en 16 endroits disposés le long de trois sections transversales. La roche a une aimantation rémanente stable, qui, d'après le test du conglomérat, devrait être originelle. Tous les endroits qui présentent des valeurs statistiques acceptables ont, sauf un, la même direction d'aimantation. Les mesures paléomagnétiques ne révèlent donc pas de structures de déformation consécutive à l'acquisition de l'aimantation rémanente. La direction moyenne de l'aimantation (11 sites, 152 échantillons) est: D = 135.9, I = —2.9 avec α_{95} = 8.8. On n'a pas apporté, pour l'instant, de correction tectonique.

Краткое содержание

Измерили природный остаточный магнетизм андезитов, покрывающих зону Sesia-Lanzo, в 16-ти точках, расположенных вдоль трех поперечных профидей. Породы характеризуются стабильным естественным остаточным магнетизмом, который, как показывают тесты на конгломератах, должен являться первичным. Во всех точках, в которых получили приемлимые статистические данные, кроме одной, установили одно и тоже направление магнетизма. Такие результаты измерения палеомганитизма указывают на то, что в последующие эпохи деформация геологических структур места не имела. В среднем направление магнетизма (11 точек, 152 пробы пород) таково: D = 135,9; ζ = −2,9 с α_{95} = 8,8. Тектонических поправок не вводили.

*) Address of the author: Roberto Lanza, Istituto di Geologia dell'Università — via Accademia delle Scienze 5 10123 Torino, Italy.

Introduction

A volcanic and detrital sequence forming the stratigraphic cover of the Sesia-Lanzo Zone (CARRARO, 1966) outcrops along over than 25 km between the Serra d'Ivrea and the Sessera Valley along the Canavese Line, an important tectonic feature separating the Sesia-Lanzo Zone, i. e. the innermost portion of the metamorphic domain of the Western Alps, from the Ivrea-Verbano Zone, i. e. a pre-alpine basement forming the outermost part of the Southern Alps.

The sequence is by no means uniform. Its lower sedimentary part is made by conglomeratic and arenitic levels devoid of volcanoclastic elements, lying on transgression on the metamorphic rocks of the Sesia-Lanzo Zone and followed by a thick conglomerate including both volcanic and metamorphic clastics. The sedimentary rocks are covered by a complex series of andesitic flows, agglomerates and tuffites. There has been much debate on the nature, structure and age of this sequence in recent years. AHRENDT (1972) suggested that the andesites cannot be regarded as the cover of the Sesia-Lanzo Zone, but constitute a type of fissure filling. CARRARO (1966, 1975) has produced data in support of the view of a synform structure. The age of the andesites – initially postulated as Carboniferous on account of the presence of plant remains (CARRARO & CHARRIER, 1972) — has now been established as Oligocene, about 30 my, by palinological and geochronological investigations (SCHEURING et al., 1974). HELLER & SCHMID (1974), in a paleomagnetic study of the Ivrea-Verbano Zone, have also examined some samples from a single site in the andesitic cover of the Sesia-Lanzo Zone.

The present paper reports the results of a 16-sites study of the remanent magnetization of the andesites, aimed to: 1) establish the mean direction of magnetization; 2) reveal any deformation that might have affected the rock after its magnetization and so altered its magnetic direction in a systematic manner.

Sampling

A geological sketch map with sampling sites is given in Fig. 1. Sampling sites lie along three sections across the regional trend of the andesites: above Favaro (FA), along the "Panoramica Zegna" road (PZ), near Piana del Ponte in the upper Sessera Valley (PP). In the FA section (BIANCHI & DAL PIAZ, 1963) the conglomerate formation is in contact with the rocks of the Sesia-Lanzo Zone and is followed by the andesites, mainly as agglomerates. A thick serpentine intercalation occurs between the volcanics and the Ivrea-Verbano rocks. In the PZ section the conglomerate level is reduced to a thickness of some metres and is followed by the andesites, mainly as flows. At the contact with the Ivrea-Verbano Zone the andesites are strongly tectonized. In the PP section the conglomerates are followed by alternate andesitic flows and tuffite levels; a sheared conglomerate level crops again out at the tectonic contact with the Ivrea-Verbano Zone (CARRARO, 1966).

The site numbers along each of these sections run progressively across the andesitic cover, from the Ivrea-Verbano to the Sesia-Lanzo Zone. At each site core samples have been taken at least one metre apart, to avoid that one clastic element only was examined when sampling the agglomerates.

Along the bed of the Elvo stream near Sordevolo, samples have been cored from six andesitic pebbles from the conglomerate formation, which is interpreted as the

result of the disgregation of early flows outpouring shortly before the main magmatic phase. These samples have been collected for the conglomerate test.

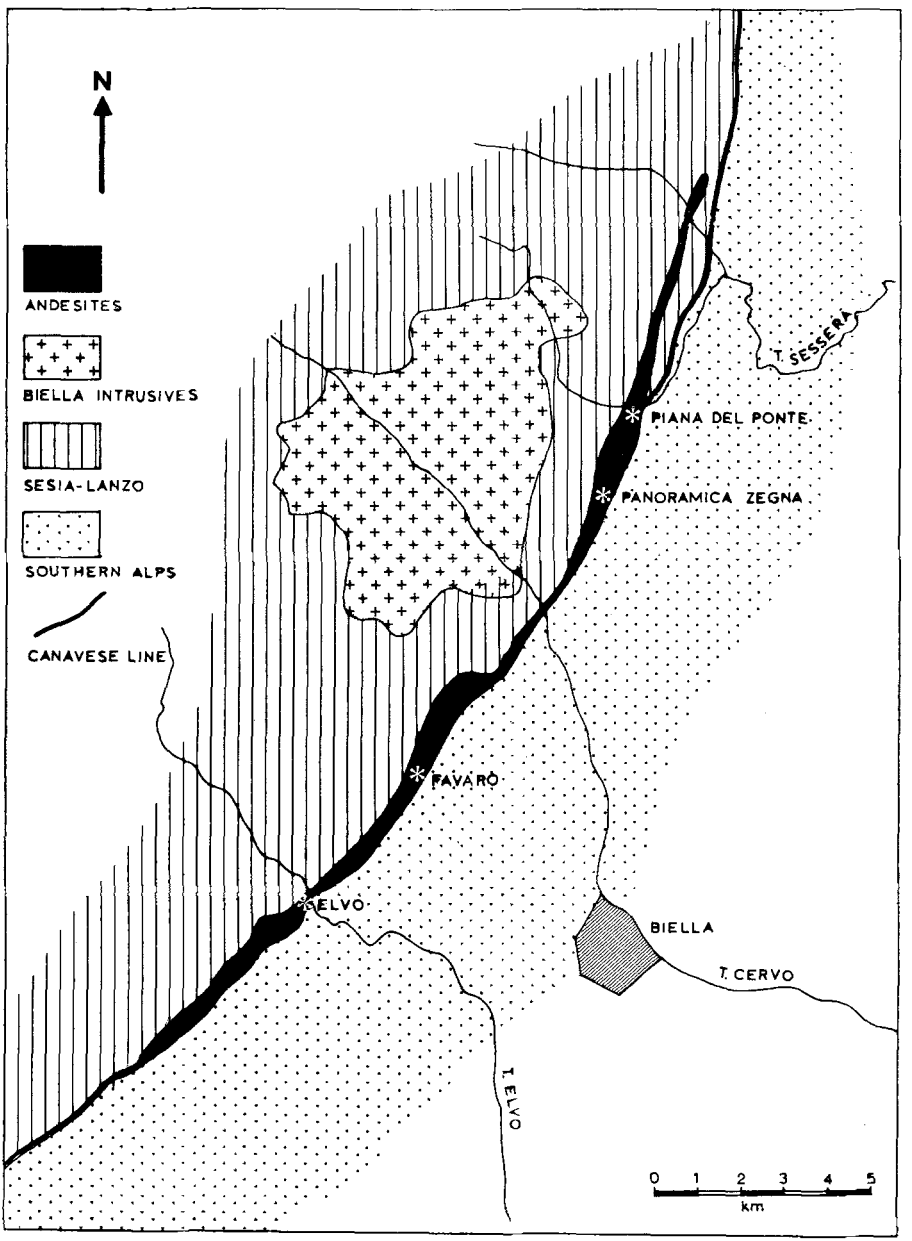


Fig. 1. Simplified geological map and sampling localities.

Core samples have been oriented both by a magnetic and a solar compass. Declination was constant within the error limits on each outcrop, ranging 1° W to 4° W in all the examined outcrops.

Measurements and results

Measurements were made by means of a SCHONSTEDT SSM-1A spinner magnetometer at the palaeomagnetic laboratory of the Laboratorio di Geologia Marina, CNR, in Bologna. Two specimens from each site have been demagnetized in alternate magnetic fields with H_{peak} intensities from 50 to 2000 Oe. Fig. 2 illustrates on an orthogonal projection (ZIJDERVELD, 1967) variations in magnetization direction and intensity during cleaning. In nearly all cases, directions of magnetization are constant within a few degrees and there is only one magnetization component. Only samples from site PZ5 display a gradual change in the inclination angle for H_{peak} values up to 300 Oe. The intensity of the natural remanent magnetization (NRM) is of the order of 10^{-3} gauss. It shows a regular decrease during cleaning in some two-thirds of the pilot specimens; in the remainder, magnetization is particularly hard and 80 % of the initial intensity is still present with $H_{peak} = 2000$ Oe.

Magnetization stability and its likely primary origin are also confirmed by the conglomerate test performed on pebbles from the Elvo outcrop. Table 1 and Fig. 3 report data on characteristic remanent magnetization (ChRM), with close clustering

Pebble	N	D	I	R	α_{95}
1	7	9.2	27.9	6.98	3.3
2	8	168.5	42.0	7.98	2.7
3	7	198.7	38.9	6.99	2.1
4	12	22.9	-17.2	11.91	3.9
5	8	275.3	53.8	7.42	16.6
6	9	139.8	-25.2	8.74	9.5

Table 1. Characteristic remanent magnetization (ChRM) of pebbles from the Elvo outcrop. N, number of specimens; D, declination; I, inclination. Statistical parameters: R, length of the resultant vector; α_{95}, angle of confidence.

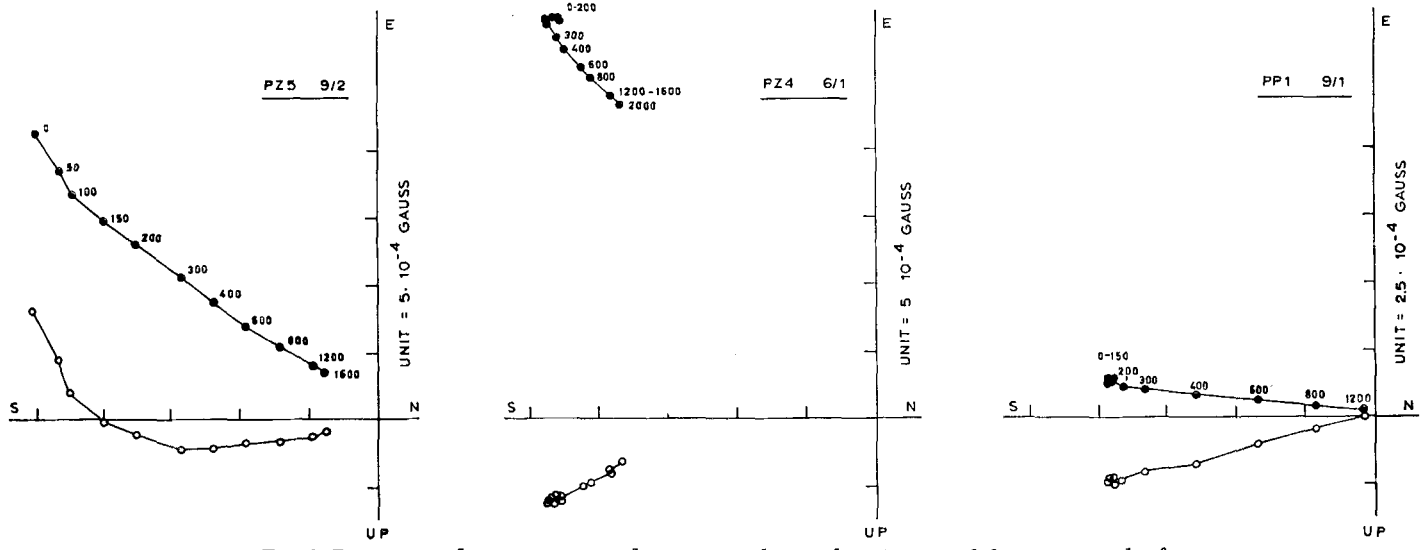


Fig. 2. Progressive demagnetization diagrams: orthogonal projection of the vector path of the NRM during AF cleaning. Solid circles: projections on the horizontal plane; open circles: projections on the vertical North-South plane. Numbers: H_{peak} values.

of directions and small α_{95} values for samples from each pebble and high scattering of mean directions of pebbles. By applying Watson's test, it is found that the length R, resultant of the standardised vectors (R = 1.93), is lower than the significance point R_o for N = 6 (R_o = 3.85). It may thus be inferred that each pebble has preserved its own ChRM direction, i. e. the magnetization is stable and of primary nature.

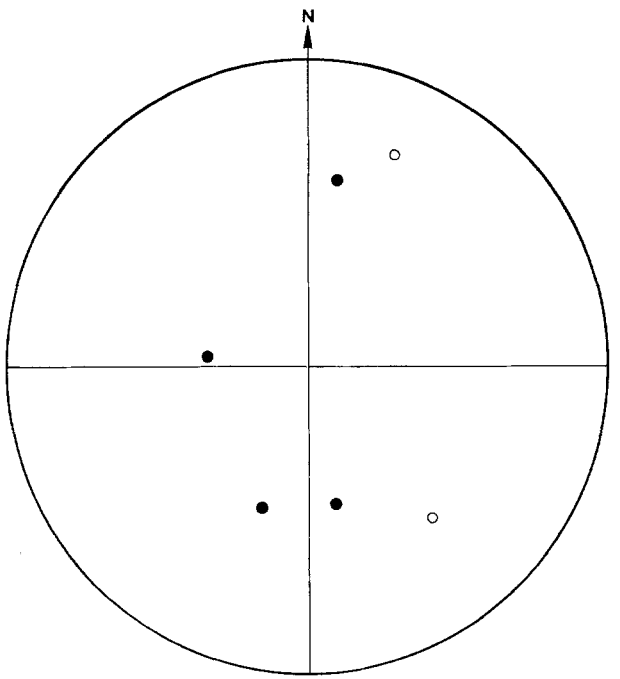


Fig. 3. Stereographic projection of the ChRM directions of the Elvo conglomerates. Solid circles: lower hemisphere; open circles: upper hemisphere.

NRM and ChRM after cleaning at the most suitable H_{peak} values have been measured for all specimens. The results are given in Table 2 and Fig. 4. The direction of magnetization is constant along PP and PZ sections: all sites give close directional values, with the exception of an anomalous inclination in the site PP4. A considerable scatter of directional values is found in the FA section, because the andesites are mostly agglomerates and because number and size of the clastic elements increase as contact with the Sesia-Lanzo Zone approaches. It is inferred that part of the cored samples were cut from agglomerate elements, hard to be discerned on the outcrop, each carrying its own magnetization direction. In the sites FA1 and FA2 (both near the Ivrea-Verbano rocks) ChRM direction coincides well with that of sections PP and PZ and the precision of results is still acceptable. In sites FA3, FA4, FA6 the mean directions are not clearly defined and the R values

		NRM		ChRM				
Site	N	D	I	H_{peak}	D	I	R	α_{95}
PP 1	14	144.8	-10.4	300	144.1	-10.0	13.33	9.2
PP 2	16	139.1	13.9	300	138.1	12.7	15.91	2.8
PP 3	15	129.6	10.7	300	129.6	10.6	14.80	4.6
PP 4	10	125.0	57.0	300	128.7	46.0	9.92	4.7
PP 5	12	135.7	0.9	300	135.8	-0.2	11.98	1.8
PZ 1	13	127.6	-28.4	300	127.3	-29.9	12.89	4.0
PZ 2	10	135.7	7.6	300	135.6	1.2	9.95	3.5
PZ 3	14	174.6	-10.1	300	135.7	4.8	13.85	4.3
PZ 4	15	134.4	-2.9	300	135.9	-5.7	14.92	3.0
PZ 5	14	121.4	32.2	200	128.0	-0.5	13.80	5.0
FA 1	16	129.7	15.1	300	135.4	9.1	14.45	12.5
FA 2	13	145.6	4.0	200	150.0	-25.4	12.69	6.8
FA 3	17	136.2	12.9	300	141.1	5.8	5.32	56.7
	(8)	(147.0)	(-13.5)	(300)	(148.3)	(-16.5)	(6.86)	(24.3)
FA 4	15	144.2	18.8	300	163.2	-4.5	5.78	51.8
	(8)	(126.2)	(13.0)	(300)	(137.3)	(0.99)	(7.22)	(19.6)
FA 5	21	60.9	48.2	300	71.9	33.2	15.93	18.5
FA 6	11	183.0	22.2	200	183.3	24.9	3.98	67.5
	(4)	(154.3)	(-22.6)	(200)	(157.4)	(-27.4)	(3.82)	(23.2)

Table 2. Natural (NRM) and characteristic (ChRM) remanent magnetization of andesites from the cover of the Sesia-Lanzo Zone. Sites: PP = Piana del Ponte; PZ = "Panoramica Zegna" road; FA = Favaro. N, number of specimens; D, declination; I, inclination; H_{peak}, peak field for AF demagnetization. Statistical parameters: R, length of the resultant vector; α_{95}, angle of confidence.

are constantly lower than the significance point R_0. Even so and in spite of the general scatter of directional values, directions not far apart from those from the other sites can be discerned. Table 2, besides the data from all the specimens collected in each of the sites FA3, FA4, FA6, gives too the data from the specimens whose direction is similar to that of sites FA1 and FA2. The ratio between significant and total samples are 8/17, 8/15, 4/11 respectively. These data show that the directions of magnetization do not undergo systematic changes even along the FA section. Nevertheless, these values are excluded from the overall mean direction of the andesitic cover, because the statistical values are poor and a large number of specimens has to be discarded. Lastly, the directions from 14 specimens from site FA5 are consistent with each other and different from the mean direction from the preceding sites, while directions from 7 specimens are scattered. The 14 specimens come from a single large clastic element whose casually oriented direction has a determining influence on the mean direction of the site.

Eleven sites have been selected for the mean direction of magnetization: 4 from the PP section (PP4 was discarded on account of its anomalous inclination); 5 from the PZ section; 2 from the FA section (FA1, FA2).

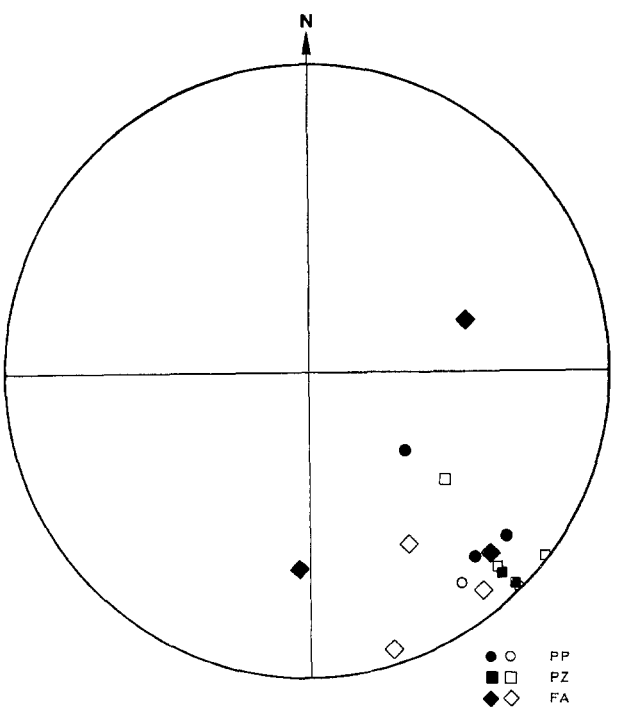


Fig. 4. Stereographic projection of the ChRM directions of the andesites from the cover of the Sesia-Lanzo Zone. Solid dots: lower hemisphere; open dots: upper hemisphere.

Discussion

ChRM direction is constant all along the andesitic cover of the Sesia-Lanzo Zone. Stability during AF cleaning and conglomerate test at one site give good evidence for a primary remanence of magnetization. No evidence for deformation subsequent to the magnetization of the rock was found. In particular, present palaeomagnetic data do not support the existence of a synform structure in the andesites.

The mean direction of magnetization in the andesites of the Sesia-Lanzo cover is: D = 135.9, I = —2.9, with N = 11, R = 10.64 and α_{95} = 8.8. The value found by HELLER & SCHMID (1974) on a single site near the PZ section is not far from this value.

The present attitude of the andesites, dipping 60°—70° toward SE, is explained either as a fissure filling or as the result of ancient subhorizontal flows. In the first case, their magnetization direction is primary and unaffected by tectonic displacement; in the second, a rotation of 60°—70° about the regional azimuth direction must be made in order to reconstruct their original attitude and direction of magnetization. The mean direction found by the second method lies close to that of the granites in the Bregaglia Massif, whose radiometric age is of about 25 my (HELLER & SCHMID, 1974; HELLER, 1973). The lack of palaeomagnetic data for the Western Alps leaves no choice between these two alternatives. Some assistance in this respect may perhaps be obtained from the NRM measurements now being made on the coeval Biella pluton, that has intruded into the Sesia-Lanzo metamorphic rocks, and is mainly composed of sienites and monzonites. Some of the mean magnetization directions so far obtained are close to those measured in the andesitic cover. It would thus seem that any rotation of the andesites would have at least partially involved the pluton as well. SCHEURING et al. (1974) have pointed out that the plutonic intrusive rocks outcrop much higher than the andesites, and explain this anomaly by postulating a relative vertical shift of at least 3.5 km. Yet this suggestion does not seem to fit the seismic (BERCKHEMER, 1968; GIESE, 1968) and magnetic data (ALBERT, 1974; LANZA, 1975) for the region. These show that, in the Biella region, the discontinuity surface, which proceeds subhorizontally to form the upper limit of the Ivrea body, has a maximum depth of the order of 2—4 km. The geophysical data do not offer any evidence of a noteworthy vertical shift on the part of this surface in the small gap between the pluton and the andesitic cover.

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